

RITESH KUMAR

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EDUCATION

Arka Jain University, Jamshedpur

B.Tech in Computer Science & Engineering — CGPA: 8.47

Aug 2022 – May 2026

Jamshedpur, Jharkhand

EXPERIENCE

TechMantra Global — Web Development Intern

May 2024 – Jul 2024

Remote — Noida

- Developed the backend for a **recipe discovery app** with **5 REST endpoints** for search, lookup, and auth (*Node.js, Express*).
- Architected a **hybrid search layer** that queries an **in-memory catalog**, merges it with the external TheMealDB API, and **caps results at 50 per query**; lookups check local data first and **fall back** to the external API.
- Authored a **data seeding script** that generates 2,500+ recipes in TheMealDB's schema with a separate ID range, so local and external results share **one response format** and never collide on lookup.
- Secured the API with **stateless JWT auth** (1-hour expiry), **bcrypt** hashing, duplicate-account checks, and email/phone login.
- Built the **React** frontend with **React Router** (7 pages), **protected routes** for dashboard and profile, and a **Context API** auth store that keeps users logged in across reloads (*React 19, Vite*).

Machine Vision & Intelligence Lab — Research Intern

May 2025 – Present

NIT Jamshedpur | *Steel defect classification on NEU-DET & SteelDefectX (1,800 + 1,631 images)* | *GitHub*

- AMFF-CNN (MobileNetV2 + FPN)**: Designed a 3.4M-parameter CNN that fuses multi-scale FPN features using channel/spatial attention and learned cross-scale weights; **92.95% macro-F1** on a leakage-free held-out split, Grad-CAM weak localization, 118 ms CPU inference; 6-variant ablation across 3 seeds (*TensorFlow*).
- TinySteelLLM (Transformer, from scratch)**: Converted ~30 handcrafted OpenCV descriptors (edges, GLCM texture, FFT, region stats) into short text prompts and classified them with a **2-layer, 4-head Transformer encoder** trained from random initialization — no pretrained weights or external APIs (*PyTorch*).
- SteelSense-BiLSTM**: Scaled this to 92 descriptors tokenized into ~94 tokens and a **2-layer BiLSTM** with attention/max/mean pooling (1.75M parameters); **99.67% ± 0.23 on NEU-DET** and **99.02% ± 0.40 on SteelDefectX** (5 seeds), benchmarked against XGBoost, Random Forest, SVM, MobileNetV3, ShuffleNetV2; first-author paper under review.

PROJECTS

QuickFix: AI-Powered Complaint Management System | Live | *GitHub*

Dec 2025

- Launched a support platform where an **async orchestrator** runs category, sentiment, and priority agents **in parallel** on each complaint, then drafts a policy-grounded reply using **RAG**; handles English, Hindi, and Hinglish (*FastAPI, React 19, SQLAlchemy, Turso*).
- Devised a **fault-tolerant 3-tier LLM fallback** — Groq LLaMA 3.3 → Gemini 2.0 Flash (multi-key rotation) → local TF-IDF/TextBlob — so complaints still get processed when a cloud API fails or hits its rate limit.
- Introduced an **in-memory cache** for repeated queries, cutting API response time from **2.5 s to ~0.5 s**; guarded the **REST APIs** with Google OAuth 2.0 + email OTP and **JWT-based RBAC**.

Community AI: Multilingual Government Scheme Assistant | Live | *GitHub*

Jan 2026

- Delivered a platform used by **100+ active users** to find **50+ government schemes**, courses, and jobs, with an AI assistant (Amazon Q → Groq → Gemini fallback) answering in **5 Indian languages** (*FastAPI, React, SQLAlchemy, Turso*).
- Created an **automated content scanner** that pulls new schemes and courses from **2 external APIs** (News, YouTube), adds them to the platform, and emails summaries via Brevo.
- Implemented **27 REST endpoints** with **async I/O**, **background tasks** for email, **connection pooling**, and indexed queries, plus JWT + bcrypt auth, Pydantic validation, and **rate limiting**.

SOEIT Student Achievement Portal | Live | *GitHub*

Feb 2026

- Shipped the achievement portal for Arka Jain University's School of Engineering & IT: **students upload certificates**, **faculty verify them**, **admins manage users** and view analytics (*Node.js, Express 5, React 19, Turso*).
- Automated **category-based point scoring** that feeds a **weekly leaderboard**, a hub of **90+ curated hackathons**, and **one-click ATS resume export** (DOCX/PDF).
- Optimized the **stateless REST backend** with **SQL indexes** for sub-100 ms lookups on 10k+ records, plus JWT-based RBAC, BLOB storage for certificates, and a React Native companion app.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C++, SQL

Backend: FastAPI, Node.js, Express.js, REST APIs, OpenAPI/Swagger, Pydantic, SQLAlchemy, JWT, OAuth 2.0, RBAC

Databases: PostgreSQL, MySQL, SQLite (Turso), MongoDB — schema design, indexing, connection pooling

System Design: Caching, Fault Tolerance & Fallbacks, Rate Limiting, Async I/O, Concurrency, Stateless Services

Frontend & DevOps: React.js, React Router, React Native, Git, GitHub Actions (CI/CD), Docker, Docker Compose, Vercel

AI / ML: PyTorch, TensorFlow, OpenCV, scikit-learn, RAG, LLM APIs

CS Fundamentals: Data Structures & Algorithms, OOP, Design Patterns, DBMS

PUBLICATIONS & ACHIEVEMENTS

Research paper: SteelSense-BiLSTM, first author with K. K. Singh — *under review, Discover Computing (Springer Nature), 2026*.

IBM Hackathon (2025): National Finalist — qualified via IBM HackerRank, competed in 24-hour national round.

AutonomousHacks '26 — GDG Gandhinagar: Offline finalist from **2,000+ participants**, Agentic AI track.

CodeFest — Arka Jain University: 2nd place, coding competition. **NPTEL:** DBMS certification.

Leadership: Gen. Secretary & President, Code & Compute Society; Community Lead, GDG on Campus; Organizer, Hack Horizon 1.0 & 2.0.